

Model Development

POS 6933 | Spring 2026

Model Development

So far, we have shown the sequential steps of the forecasting exercise. Specifically, we have defined the outcome of interest as log fatalities across seven future forecast horizons, specified a partition strategy to divide the data into training and test sets, addressed missingness through imputation, and developed two benchmark models for evaluation purpose. Most importantly, in week 8, we show in a tutorial how to develop machine learning models on a sample of predictors and evaluate their performance against the two benchmark models.

In the remaining weeks of the semester, your task is to design, develop, and evaluate your own theory-driven forecasting models. In other words, you will develop a theoretical model using two selected algorithms, and compare their forecasting performance against the two benchmark models developed earlier in the course (in week 5). The following passages show how you should approach this task.

Theoretical Models

In field of the conflict studies, scholars have developed a wide range of theoretical models to explain conflict onset and other forms of political violence. These frameworks typically emphasize a range of indicators such as state weakness, grievances and exclusion, mobilizational capacity, regime type, etc... Your task is to pick a theoretical model listed below using the indicators (variables) exist in the imputed dataset. Once you choose a theoretical model, then you need to identify the most appropriate indicators already available in the dataset, justify (based on existing literature) why those variables correspond to that theory, and group them into a coherent modeling block. You may pick one of following theoretical models:

1. **Political Opportunity Model:** Focus on political openings and constraints in the political environment.
2. **Grievance Model:** Capture deprivation, injustice, exclusion, discrimination, and perceived unfairness.
3. **Resource Mobilization Model:** Measure access to material, organizational, geographic, and social resources.
4. **Democratization / Regime Transition Model:** Assess periods of regime transition, liberalization, or institutional change.
5. **Modernization Model:** Examine the impacts of social and economic transformation, urbanization, education, and structural change.
6. **Regime Characteristics Model:** Capture the characteristics of the regime itself, such as authoritarianism, hybrid rule, or institutional weakness.
7. **State Capacity Model:** Focus on the state's administrative reach, ability, extraction capacity, and ability to enforce order.
8. **Polarization Model:** Focus on ethnic or demographic characteristics of the country such as , ethnic or sectarian fragmentation, or polarization.
9. **Repression Model:** Measure coercion, repression, human rights abuse, arrests, or prior state violence
10. **Political Shock Model:** Capture abrupt shocks such as coups, invasions, regime collapse, foreign power withdrawal, or elite crises.
11. **Natural Resources Model:**Capture lootable resources, rents, extractive wealth, or economic structures.

Algorithms

After choosing the theoretical model and selecting the appropriate and theoretically justified indicators that capture its core dimensions, the next step is to fit that theory using two machine learning algorithms from the list below. They are:

- Random Forest
- Gradient Boosting Machine

- Neural Network

You may find that several of these algorithms have variants or modified implementations. If you choose one of these variants rather than the standard form of the algorithm, justify that choice and provide a brief explanation of why it is more appropriate.

Model Evaluation

After estimating your models, the next step is to evaluate their out-of-sample predictive performance against the two benchmark models developed in Week 5. Model comparison must be based on standard predictive accuracy metrics, including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the Continuous Ranked Probability Score (CRPS). CRPS allows you to assess not only how close predictions are to the observed outcomes on average, but also how well the models quantify predictive uncertainty.

The purpose of this assignment is to determine whether your theory-driven machine-learning model improve upon the baseline benchmarks in terms of predictive accuracy and probabilistic performance.

Final Product

After completing the three stages above, the final deliverable must be a professional analytical memo of approximately 7–10 pages in a research paper format. The memo should clearly present the full modeling strategy, from theoretical selection to predictive evaluation, in a way that is both methodologically and substantively clear. Table 1 presents the main components of the memo.

Table 1: Memo Components

Component	Description
Select your theory	Identify the theoretical model and explain its relevance for conflict forecasting.
Operationalize your theory	Describe how the theory is operationalized, including the indicators used and the total number of indicators in the specification.
Fit your theory	Fit and explain which algorithms were selected, why they were chosen, and how they work.
Evaluate your model	Compare predictive power of your theory to the Week 5 benchmark models using clear tables and publication-quality plots based on MSE, RMSE, and CRPS across all time horizons.

The memo should then conclude with a substantive discussion of the findings. This section should discuss whether your theoretical model is consistently stronger across algorithms and time horizons, and what the results imply for conflict forecasting more broadly.